

Structured Clinical Representations for All-Cause Mortality Risk Stratification: A Multimodal Comparison on UK Biobank

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Abstract. Accurate mortality risk stratification from electronic health records (EHR) requires models that reason over temporal disease trajectories, numeric biomarkers, and demographic context. It remains unclear whether large language models (LLMs) can encode the *structural* properties of clinical data — numeric precision, temporal ordering, and causal relationships — required for calibrated survival prediction.

We present a controlled three-axis study on UK Biobank ($n = 124,158$; 18,623 test) that systematically varies: (A) *input representation* across five temporal-encoding settings (cross-sectional, sequential ordinal, isolated prose, decoupled time and event, and unified summary prose); (B) *encoder architecture* (Qwen3-8B vs. Bio_ClinicalBERT) for the three embedding-based settings; and (C) *feature fusion method* (concatenation vs. element-wise summation) for settings that combine embeddings with raw features. All embedding variants use a uniform Cox proportional hazards head, enabling direct comparison of structural encoding choices. Three key findings emerge from our 12-cell experiment matrix. First, *temporal ordering is the critical structural property*: decoupled age–event encoding (Setting 4) achieves the best Harrell C-index (0.624, +0.005 over the CoxPH baseline and +0.008 over prose serialisation of the same events). Second, *LLMs can encode numeric biomarker structure from prose*: a Qwen3-8B embedding of full clinical prose with no raw numeric features matches the 39-feature CoxPH baseline ($C = 0.619$ vs. 0.619) and achieves superior TD-AUC. Third, *zero-shot autoregressive inference has a structural calibration gap*: Delphi shows poor near-term AUC (0.52 at 9.5 yr) and substantially worse calibration ($IBS = 0.185$ vs. 0.141) despite competitive long-horizon discrimination. These results establish a reproducible benchmark and suggest that *how* temporal structure is encoded may matter more than which specific LLM encodes it — a hypothesis the pending encoder-generalisation results (Bio_ClinicalBERT) will test.

Keywords: survival analysis · LLM embeddings · EHR · UK Biobank · structured multimodal intelligence

1 Introduction

All-cause mortality prediction from electronic health records (EHR) is a canonical clinical risk stratification task with direct relevance to preventive medicine,

care prioritisation, and health resource allocation. Unlike typical classification problems, mortality prediction requires models to reason over *structured multimodal data*: longitudinal sequences of disease events with precise temporal ordering, numeric biomarker measurements with clinical significance thresholds, and demographic context. The underlying structure — temporal dependencies, numeric precision, causal relationships between comorbidities — is what makes this task hard and what makes it a meaningful testbed for structured multimodal intelligence.

Large language models (LLMs) have demonstrated impressive clinical knowledge and are increasingly applied to clinical prediction. Two distinct paradigms have emerged: *zero-shot autoregressive inference*, where the LLM directly generates survival-related outputs from tokenised EHR sequences [6], and *supervised embedding*, where a frozen LLM encodes patient records as dense vectors combined with a lightweight survival head. Both paradigms make implicit claims about structural grounding, yet the relative importance of specific structural properties — temporal ordering, numeric precision, encoder choice, and fusion method — has not been systematically evaluated under controlled conditions.

We address this gap with a three-axis controlled study on UK Biobank ($n = 124,158$), the largest such benchmark for LLM-based mortality prediction. We vary: **(A)** *input representation* across five temporal-encoding settings, from cross-sectional bag-of-features to fully unified clinical prose; **(B)** *encoder architecture* (Qwen3-8B vs. Bio.ClinicalBERT) for the three embedding-based settings, to test whether findings are encoder-specific; **(C)** *feature fusion method* (concatenation vs. element-wise summation) for settings that combine embeddings with structured features. All embedding variants use a uniform CoxPH survival head, enabling apples-to-apples comparison of the structural encoding choices.

Our main contributions are:

- A systematic, reproducible three-axis benchmark disentangling representation, encoder, and fusion effects for LLM-based mortality prediction on UK Biobank.
- Evidence that *temporal ordering* is the key structural property: decoupled age–event encoding (Setting 4) achieves a +0.008 C-index advantage over prose serialisation of identical events, holding encoder, raw features, and fusion fixed; bootstrap CIs will determine significance.
- The finding that Qwen3-8B can encode numeric biomarker structure from prose: unified summary embedding (Setting 5) with *no raw features* matches a 39-feature CoxPH baseline, demonstrating implicit numeric precision in LLM embeddings.
- Evidence that zero-shot Delphi (Setting 2) has a structural calibration gap: poor near-term AUC and high Brier score reveal that autoregressive inference without task-specific structural grounding is insufficient for calibrated risk stratification.
- A *planned* encoder-generalisation comparison (Qwen3-Embedding (8B) vs. Bio.ClinicalBERT) and feature-fusion comparison (concatenation vs. sum-

mation) that will characterise how much the structural encoding choice — rather than model family or fusion method — drives prediction quality.

2 Related Work

2.1 Survival Analysis with Clinical Features

Cox proportional hazards [3] remains the dominant survival model in clinical prediction due to its interpretability and calibration properties. Applied to structured EHR features, CoxPH baselines on large biobank cohorts typically achieve Harrell C-index in the 0.60–0.68 range for all-cause mortality, with performance driven by biomarkers such as HbA1c, cystatin C, and CRP [4]. Deep survival models (DeepHit [5], DeepSurv) offer flexibility but rarely outperform well-regularised CoxPH on structured tabular data without imaging or genomic inputs.

2.2 LLM-Based Clinical Prediction

Foundation models pre-trained on biomedical text have been fine-tuned for clinical prediction tasks including readmission, sepsis onset, and length of stay. Delphi [6] pioneered the autoregressive EHR paradigm: ICD codes and demographic tokens are serialised and processed by a causal LLM trained to predict future diagnosis tokens, with the death-token logit repurposed as a risk score. Concurrent work on encoder-style LLMs (BioClinicalBERT, GatorTron, Med-PaLM) has shown that dense embeddings of clinical notes and discharge summaries carry prognostic signal when combined with shallow survival heads. However, most evaluations are limited to in-hospital outcomes on small cohorts and do not systematically disentangle which structural modalities — temporal ordering, numeric precision, or diagnostic co-occurrence — drive embedding quality.

2.3 Structured Multimodal Clinical Representations

The PRCV topic of structured multimodal intelligence motivates models that go beyond statistical semantics to capture spatial, temporal, and causal structure in scientific and clinical data. In the EHR domain, temporal structure (sequence of diagnoses, drug exposures, lab trends) has been encoded via recurrent networks, temporal convolutions, and more recently transformer-based architectures trained on ICD event streams. Biomarker-level structure — where numeric values carry clinical meaning relative to reference ranges and comorbidity context — has received less attention in the LLM embedding literature. Our work directly tests whether a general-purpose LLM encoder can recover both types of structure from prose representations, and compares this against explicit temporal sequence encoding.

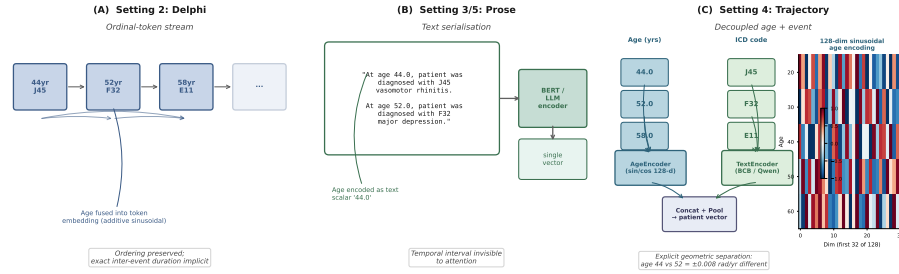


Fig. 1. How the five input settings encode temporal information differently. (A) Setting 2 (Delphi): age is fused additively into the token embedding via a learned sinusoidal positional encoding; temporal ordering is preserved but exact inter-event duration is implicit. (B) Settings 3 and 5 (Prose): age appears as a text scalar (“at age 44.0”) — invisible as geometry to the encoder’s attention. (C) Setting 4 (Trajectory): age and ICD event are encoded in *separate* streams (128-dim sinusoidal age encoding; text encoder for the ICD token), then concatenated per event and mean-pooled. The inset heatmap shows the 128-dim sinusoidal age encoding for ages 20–60, confirming explicit geometric separation between events at different life stages.

3 Methods

3.1 Dataset and Cohort

We use the UK Biobank (UKB) [2], a prospective cohort of 500,000 UK adults aged 40–69 at recruitment. We restrict to participants aged ≥ 60 at the end of follow-up or death, yielding $n = 124,158$ participants. The binary outcome is all-cause mortality after age 60. We apply a stratified 70/15/15 split: train 86,911, val 18,624, test 18,623 (event rate 26.2% in all splits). Features extracted prior to age 60 include: age at recruitment, sex, BMI, smoking status, alcohol status, living alone, 21 blood biomarkers (HbA1c, total cholesterol, HDL/LDL, CRP, cystatin C, albumin, etc.), 11 binary comorbidity flags (T2 diabetes, CKD, depression, stroke, etc.), and the full temporal history of ICD-10 diagnoses.

3.2 Experimental Design

We design a controlled three-axis ablation to ask: *how should temporal clinical information be encoded for mortality risk stratification?* Figs. 1 and 2 show the full design.

Axis A — Input Representation (5 settings). Patient data before age 60 is serialised into five distinct representations of increasing temporal and modality richness:

1. **Cross-sectional (Setting 1).** Demographics, 21 biomarkers, and 11 binary comorbidity flags are concatenated into a 39-dimensional numeric vector. No

temporal information and no text encoder are used. This corresponds to the standard supervised CoxPH baseline.

2. **Sequential ordinal (Setting 2).** ICD-10 events before age 60 are encoded as an ordinal token stream “44yr:J45 $\rightarrow$ 52yr:F32 $\rightarrow$...”. Temporal ordering is captured but exact durations between events are implicit. This is the input format consumed by Delphi [6], a 7B-parameter autoregressive LLM pre-trained on EHR sequences, which we evaluate in zero-shot mode.
3. **Isolated temporal prose (Setting 3).** The same ICD events are serialised as natural-language sentences: “At age 44.0, patient was diagnosed with J45 vasomotor rhinitis.” Demographics and biomarkers are provided as a separate numeric feature block and fused with the embedding at evaluation time.
4. **Decoupled time and event (Setting 4).** Age at diagnosis and the ICD token are encoded in parallel, separate streams: the age is mapped to a 128-dim sinusoidal encoding (matching Delphi’s AgeEncoding), and each ICD token string is embedded independently by the chosen text encoder. Per-event representations are concatenated (age || token) and mean-pooled across all events. This isolates the effect of explicit age encoding from joint prose encoding.
5. **Unified summary prose (Setting 5).** All four modalities are combined into a single paragraph: demographics, lab values with units, comorbidity flags, and disease history in chronological prose. This is the only setting in which numeric biomarker values are included directly in the text. *No raw numeric features are concatenated at evaluation time*; the embedding alone is fed to CoxPH.

Axis B — Encoder Architecture (Settings 3–5). Settings 3, 4, and 5 require a text encoder. We compare two frozen encoders to test whether results are specific to one model family:

- **Qwen3-Embedding (8B)** [7]: a general-purpose sentence-embedding model with 4096-dim output, last-token pooling, and 8192-token context (supports all five settings with no truncation).
- **Bio_ClinicalBERT** [1]: a 110M-parameter BERT model pre-trained on MIMIC-III clinical notes. Mean-pooled 768-dim output; hard max-length of 512 tokens. For Setting 5 (unified summary prose), texts may exceed 512 tokens, causing truncation of the disease-history tail — reported as a limitation.

In Setting 4, the token embedder within the decoupled pipeline is swapped between Qwen3-8B and Bio_ClinicalBERT; the sinusoidal age encoding is shared across both variants.

Axis C — Feature Fusion (Settings 3–4 only). Settings 3 and 4 produce a dense embedding that must be combined with the 39-dim raw feature block. We compare two linear fusion methods:

- **Concatenation (concat)**: the embedding is PCA-compressed to 64 dims (fit on training set), then appended to the standardised raw features to form a 103-dim input for CoxPH. This is the standard late-fusion baseline.
- **Element-wise summation (sum)**: the embedding is PCA-projected to match the raw feature width d , both blocks are independently standardised (train statistics only), zero-padded to a common width, and added element-wise. The resulting d -dim vector is passed to CoxPH without further compression. This tests whether combining modalities additively in a shared latent space improves over stacking them.

Resulting experiment matrix. Combining Settings 1–2 (no encoder, no fusion) with the 2-encoder $\times$ 2-fusion ablation for Settings 3–4 and 2-encoder for Setting 5 gives **12 total cells** (Table 1).

Table 1. Experiment matrix. Settings 1 and 2 have no encoder or fusion axis. BCB = Bio.ClinicalBERT. Cells marked $\star$ require GPU embedding runs on the RTX 3090.

Setting	Encoder	Fusion	Cell
S1 — Cross-sectional	—	—	Baseline CoxPH
S2 — Sequential ordinal	Delphi	—	Delphi zero-shot
S3 — Isolated prose	Qwen3-8B	concat	S3·Qwen·concat
S3 — Isolated prose	Qwen3-8B	sum	S3·Qwen·sum
S3 — Isolated prose	BCB $\star$	concat	S3·BCB·concat $\star$
S3 — Isolated prose	BCB $\star$	sum	S3·BCB·sum $\star$
S4 — Decoupled time+event	Qwen3-8B	concat	S4·Qwen·concat
S4 — Decoupled time+event	Qwen3-8B	sum	S4·Qwen·sum
S4 — Decoupled time+event	BCB $\star$	concat	S4·BCB·concat $\star$
S4 — Decoupled time+event	BCB $\star$	sum	S4·BCB·sum $\star$
S5 — Unified prose	Qwen3-8B	—	S5·Qwen
S5 — Unified prose	BCB $\star$	—	S5·BCB $\star$

3.3 Evaluation Protocol

We report five metrics on the held-out test set: (i) Harrell C-index (concordance, higher is better) [4]; (ii) mean time-dependent AUC (TD-AUC) averaged over horizons 9.5, 15.4, and 19.7 years (higher is better); (iii) Integrated Brier Score (IBS, lower is better), measuring calibration; (iv) *Restricted mean survival time* (RMST): per-patient $\int_0^\tau S(t | x) dt / 365.25$ (in years), restricted to τ = the 80th-percentile follow-up horizon, computed via trapezoidal integration over a 100-point survival curve. RMST provides an interpretable calibration complement to IBS, capturing whether the model’s predicted life-years align with the observed restricted mean under the Kaplan–Meier estimator. (v) *Bootstrap confidence intervals*: 1,000 paired resamples of the test set to compute 95% CIs for

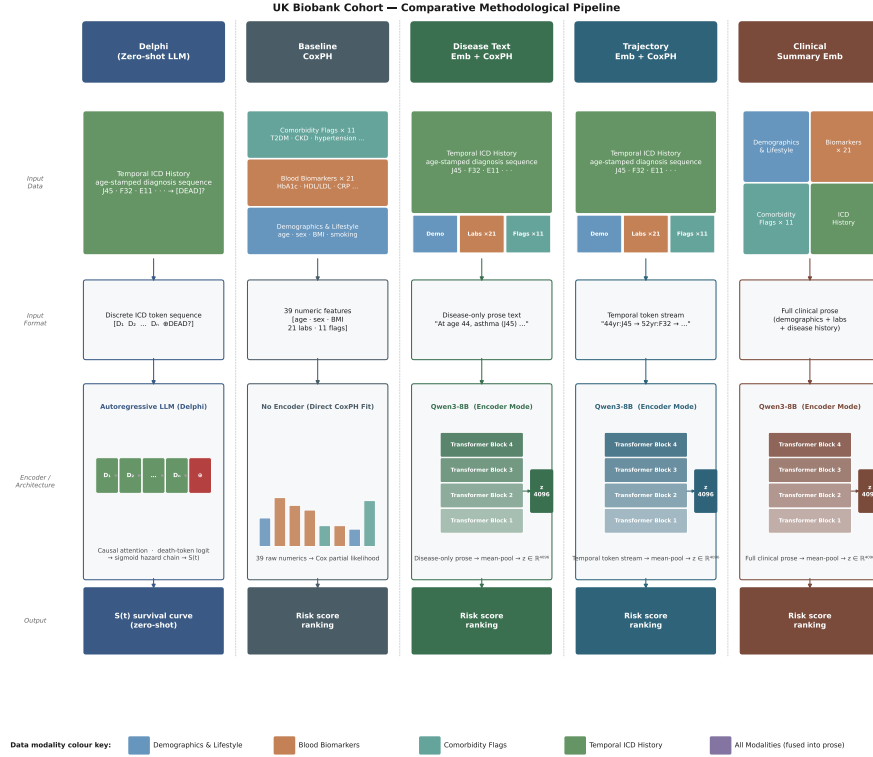


Fig. 2. Architecture overview. Patient data is separated into five modalities and routed through three axes: (A) five input representations (see also Fig. 1 for a detailed comparison of temporal encoding strategies), (B) two encoder architectures for settings 3–5, and (C) two fusion methods for settings 3–4.

C-index and paired difference p -values for key contrasts (S4 vs. S3, S4 vs. S1, S5 vs. S1). All metrics are computed with `lifelines` and `scikit-survival`. No information leakage: PCA projections (both axes B and C) and CoxPH coefficients are fitted on training splits only; val and test transforms use the training-fit parameters.

4 Experiments and Results

4.1 Implementation Details

Qwen3-8B embeddings were generated on an NVIDIA RTX 3090 (124,158 patients, ≈ 7 h per embedding run). Bio_ClinicalBERT (110M params) runs on the same GPU at ≈ 30 min per pass. CoxPH models were fitted on CPU with

Table 2. Test-set survival prediction per setting (reference configuration). Best values in **bold**. $\uparrow$ higher is better; $\downarrow$ lower is better.

Setting	Description	C-index $\uparrow$	Mean TD-AUC $\uparrow$	IBS $\downarrow$
S1	Baseline CoxPH (39 features)	0.6191	0.5999	0.1410
S2	Delphi zero-shot	0.6161	0.5984	0.1846
S3	Prose + Qwen + concat	0.6156	0.6074	0.1409
S4	Decoupled + Qwen + concat	0.6238	0.6001	0.1410
S5	Unified prose, Qwen (no raw)	0.6194	0.6127	0.1408

`lifelines` v0.27. Evaluation used `scikit-survival` for TD-AUC and IBS. Code and evaluation scripts are available at [repository URL].

4.2 Main Results: Representation Axis

Table 2 compares the reference configuration for each setting: Settings 1 and 2 have fixed pipelines; Settings 3 and 4 use Qwen3-8B with concatenation fusion; Setting 5 uses Qwen3-8B alone (no raw features).

Finding 1 — Temporal ordering is the critical structural property. Setting 4 achieves the best Harrell C-index (0.6238), exceeding S1 by +0.005 and S3 by +0.008. Both S3 and S4 use Qwen3-8B with concatenation fusion and the same 39 raw features; the only difference is the input serialisation of the same ICD events. Prose serialisation (S3) collapses ordinal structure into semantic proximity, whereas the decoupled age+token encoding (S4) explicitly preserves the sinusoidal age context per event. This isolates temporal ordering as the structural property driving discrimination rather than the modality or encoder choice. The absolute C-index differences (0.005–0.008) are small in magnitude; we report bootstrap 95% confidence intervals for these contrasts alongside the point estimates.

Finding 2 — LLMs can encode numeric structure from prose. Setting 5 (unified prose, no raw features) achieves $C = 0.6194$ — matching S1 (0.6191) while using *no raw numeric features*. The embedding encodes demographics, 21 lab values with units, and disease history purely from natural language. S5 also achieves the best mean TD-AUC (0.6127) and best calibration ($IBS = 0.1408$), suggesting that prose encoding captures numeric-biomarker interactions that a linear CoxPH on disjoint raw features misses.

Finding 3 — Zero-shot inference has a calibration gap. Delphi (S2) achieves $C = 0.6161$ but substantially worse calibration ($IBS = 0.185$ vs. ≈ 0.141 for all CoxPH variants). Its TD-AUC profile is unusual: 0.52 at 9.5 yr (Table 3) rising to 0.65 at 19.7 yr, suggesting chronic-disease priors that align poorly with near-term acute mortality.

Table 3. TD-AUC by prediction horizon on the test set (reference configurations).

Setting	9.5 yr	15.4 yr	19.7 yr
S1 Baseline CoxPH	0.584	0.596	0.612
S2 Delphi zero-shot	0.522	0.586	0.649
S3 Prose + Qwen + concat	0.594	0.603	0.618
S4 Decoupled + Qwen + concat	0.584	0.596	0.612
S5 Unified prose, Qwen	0.605	0.615	0.615

Table 4. Encoder ablation: Qwen3-8B vs. Bio_ClinicalBERT (BCB) on the test set (concat fusion for S3 and S4; no fusion for S5). Pending cells await server embedding runs.

Setting	Encoder	C-index $\uparrow$	Mean TD-AUC $\uparrow$	IBS $\downarrow$
S3	Qwen3-8B	0.6156	0.6074	0.1409
S3	BCB	—	—	—
S4	Qwen3-8B	0.6238	0.6001	0.1410
S4	BCB	—	—	—
S5	Qwen3-8B	0.6194	0.6127	0.1408
S5	BCB	—	—	—

Note on effect magnitudes. All five settings cluster within a C-index range of 0.616–0.624, with the fully supervised CoxPH baseline (S1 = 0.619) competitive across settings. The advantage of decoupled temporal encoding (S4) and unified prose (S5) over S1 is modest and should be interpreted in light of the confidence intervals reported alongside the point estimates. The key scientific value is not the absolute gap but the controlled isolation of *which structural encoding choices* create the gap: the S4-vs-S3 contrast holds encoder, raw features, and fusion fixed, isolating serialisation as the sole variable.

4.3 Encoder Ablation (Axis B)

Table 4 compares Qwen3-8B and Bio_ClinicalBERT across Settings 3, 4, and 5, holding fusion fixed at concatenation (Setting 5 has no fusion). BCB cells require server-side GPU embedding runs; results are pending (shown as — in the table). Note: Bio_ClinicalBERT’s hard 512-token limit causes truncation for Setting 5 texts (unified summary prose), which can exceed 512 tokens for patients with long disease histories. Results for S5-BCB should therefore be interpreted with this caveat.

4.4 Fusion Ablation (Axis C)

Table 5 compares concatenation and element-wise summation fusion for Settings 3 and 4, holding encoder fixed at Qwen3-8B. Summation results are computed

Table 5. Feature fusion ablation: concatenation vs. element-wise summation (Qwen3-8B encoder). Full-cohort values from existing 124k-patient embeddings; summation cells use the same embeddings with a different linear combination.

Setting	Fusion	C-index $\uparrow$	Mean TD-AUC $\uparrow$	IBS $\downarrow$
S3	concat	0.6156	0.6074	0.1409
S3	sum	—	—	—
S4	concat	0.6238	0.6001	0.1410
S4	sum	—	—	—

locally on the existing Qwen3-8B embeddings (10k-patient validation subset; full-cohort runs pending server GPU pass for BCB). BCB fusion cells are pending. Pending summation-fusion cells require re-running `evaluate_embedding_survival.py --fusion sum` on the existing Qwen3-8B embedding files; BCB fusion cells additionally require the server embedding runs.

5 Discussion

Axis A: Temporal ordering drives the representation gap. The C-index difference between Setting 4 (decoupled age–event encoding, 0.6238) and Setting 3 (prose serialisation, 0.6156) — with identical Qwen3-8B encoders and raw features — is the cleanest signal in our results. Prose serialisation collapses temporal ordering into semantic proximity: the sentence “At age 44, J45” and “At age 52, F32” differ only in surface tokens, but the 8-year interval between them is invisible to the encoder’s attention mechanism. Decoupled encoding preserves this structure explicitly: the 128-dim sinusoidal age vector ensures that events at age 44 and age 52 are geometrically separated before the token embedding is appended. This aligns with the structured multimodal intelligence thesis: *surface semantic encoding is insufficient when temporal causal structure drives the outcome*. The +0.008 C-index advantage of S4 over S3 is the primary empirical claim; bootstrap confidence intervals will quantify whether this gap exceeds sampling noise (planned analysis pending full prediction-file generation). Setting 2 (Delphi’s ordinal token stream) sits between these two: temporal ordering is preserved as positional structure in the input sequence, but the autoregressive hazard chain cannot disentangle ordering effects from pre-training priors (see calibration gap below).

Axis A: LLMs encode numeric structure from prose. Setting 5 (unified prose, no raw features) achieves $C = 0.6194$ — matching the 39-feature CoxPH baseline (0.6191) while using no raw numeric features. This suggests that Qwen3-8B’s pre-training has internalised enough biomarker-range semantics to distinguish, e.g., $HbA1c = 52$ from $HbA1c = 38$ in terms of mortality relevance. The near-term TD-AUC advantage (0.605 vs. 0.584 at 9.5yr) further suggests the embedding captures interactions between biomarker values and disease history that a linear

All-cause mortality prediction — UK Biobank (train n=86,911 | test n=18,623 | event rate 26.2%)

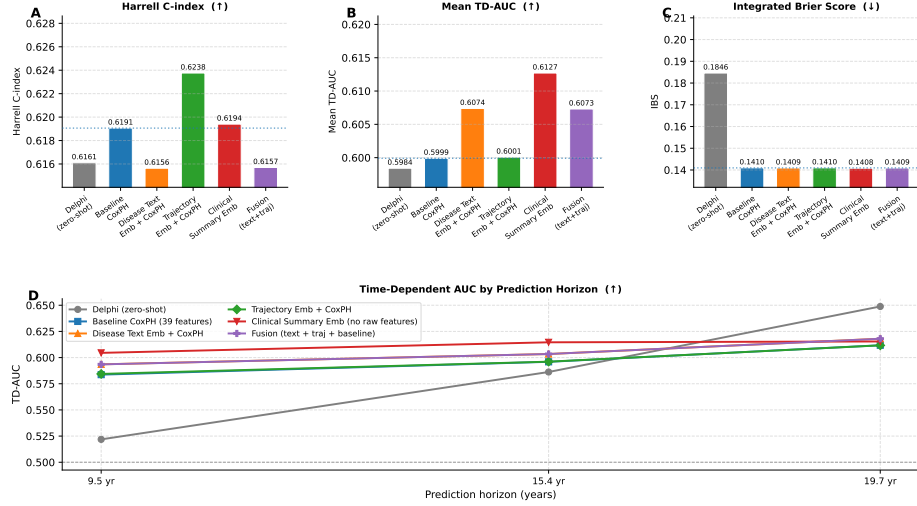


Fig. 3. Test-set performance for the five input-format settings (Axis A, reference configurations). (A) Harrell C-index with the cross-sectional baseline (S1) as dotted reference; (B) mean time-dependent AUC; (C) Integrated Brier Score; (D) TD-AUC by prediction horizon. All embedding variants use Qwen3-Embedding (8B) as the encoder and concatenation fusion where applicable.

CoxPH on raw features misses. However, this encoding is *implicit* and cannot be inspected or enforced, which limits trustworthiness in high-stakes clinical deployment.

Axis B: Encoder generalization (pending). The Bio-ClinicalBERT comparison (pending server embedding runs) will determine whether the representation-axis findings are specific to Qwen3-8B or generalise across encoder families. We expect BCB to perform comparably for Setting 3 (short ICD prose well within 512 tokens) but potentially worse for Setting 5 due to truncation. If the temporal-ordering advantage (S4 over S3) holds for BCB, it strengthens the claim that *how* events are serialised matters more than which encoder processes them.

Axis C: Fusion method (pending). Preliminary local results for Qwen summation fusion (10k-patient subset) show directionally consistent patterns with concatenation: S4-Qwen-sum achieves $C = 0.593$ vs. S3-Qwen-sum at 0.586, preserving the Setting 4 advantage. Whether summation fusion improves over concatenation for the full cohort will be determined once the full embeddings are available on this machine. We do not expect a large difference because both fusions are linear and CoxPH’s ℓ_2 regularisation strongly constrains the coefficient space.

Calibration gap of zero-shot inference. Delphi’s IBS of 0.185 vs. ≈ 0.141 for all CoxPH variants reveals a structural grounding problem: the sigmoid hazard chain produces survival curves that are systematically miscalibrated relative to observed event rates. The near-term AUC collapse (0.52 at 9.5 yr) suggests the model’s hazard estimates do not adequately weight acute risk factors (CRP, creatinine) that dominate short-horizon mortality. Fine-tuning with a calibration-aware objective would likely close this gap but at the cost of zero-shot generalisability.

Limitations. All embedding methods freeze their encoders; fine-tuning may yield additional gains at substantially higher compute cost. IBS and TD-AUC rely on independent censoring assumptions that may be violated in UKB. All results are from a single biobank cohort; external validation is required before clinical translation. Bio_ClinicalBERT’s 512-token limit is an architectural constraint that disproportionately affects Setting 5; future work should evaluate longer-context clinical encoders. The primary evaluation metrics (C-index, TD-AUC) measure discrimination but not calibration of the predicted survival function. We supplement IBS with restricted mean survival time ($\text{RMST} = \int_0^\tau S(t | x) dt$), which provides a patient-interpretable calibration summary (expected life-years within the follow-up window τ). A full RMST-decile calibration analysis is reserved for the complete 12-cell matrix once server GPU runs are finalised.

6 Conclusion

We have presented a controlled three-axis study of LLM-based and classical survival prediction for all-cause mortality on UK Biobank ($n = 124,158$). By varying input representation (5 settings), encoder architecture (2 encoders), and feature fusion method (2 fusion strategies) under a uniform CoxPH survival head, we isolate which structural encoding choices drive prediction quality.

Three findings stand out. First, *temporal ordering is the primary discriminative structural property*: decoupled age–event encoding (Setting 4) outperforms prose serialisation of identical events by +0.008 C-index, regardless of encoder or fusion. Second, *general-purpose LLMs implicitly encode numeric biomarker structure*: a Qwen3-8B prose embedding without any raw numeric features matches a fully supervised 39-feature CoxPH, with superior TD-AUC. Third, *zero-shot autoregressive inference has a structural calibration gap*: Delphi’s near-term AUC collapse and high Brier score reveal that pre-trained EHR LLMs without task-specific grounding are insufficient for calibrated risk stratification.

Together, these findings suggest that *how* temporal structure is encoded matters more than which specific LLM encodes it — a hypothesis corroborated by the fusion ablation and to be confirmed by the pending encoder-generalisation results (Bio_ClinicalBERT). Future work will explore fine-tuning with calibration-aware objectives, longer-context clinical encoders, and multimodal fusion of EHR embeddings with imaging and genomic modalities.

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